25900245 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEBTS200 - CLINICAL BIOCHEMISTRY AND CLINICAL MICROBIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 2 Hours **Maximum Marks: 70**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. Body Mass Index (BMI) above _____ indicates obesity. [K] / [CO1]
2. Common cold is primarily transmitted through _____ droplets. [U] / [CO5]
3. _____ smear is the gold standard method for malaria diagnosis. [K] / [CO5]
4. The _____ is responsible for balance, posture, and coordination of movements. [K] / [CO2, CO3]
5. In the Jaffe's reaction, creatinine reacts with _____ in alkaline medium to form a red-orange color. [U] / [CO3]
6. The_____-receives oxygenated blood from the lungs. [K] / [CO2, CO3]

7. The kidney converts inactive vitamin D into its active form _____. [A] / [CO2, CO3]

8. Amoebiasis is more common in regions with poor _____ and sanitation. [K] / [CO5]

9. Salmonella Typhimurium and Salmonella Enteritidis commonly cause _____ infection. [K] / [CO5]

10. A common biochemical marker of lung injury is increased level of _____ in serum [U] / [CO2, CO3]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **3** marks

11. State any three major lifestyle factors that contribute to kidney damage. [K] / [CO1]

12. In what way can aflatoxin damage liver cells? [U] / [CO1]

13. what are the blood tests useful in diagnosing acute pancreatitis. [U] / [CO2, CO3]

14. List two diagnostic tests used for coronavirus infection. [K] / [CO4]

15. How does alcohol affect cardiovascular health? [U] / [CO1]

16. Name two laboratory diagnostic methods for filariasis. [U] / [CO5]

17. State the role of the liver in detoxification with an example. [U] / [CO2, CO3]

18. Mention the normal range of total cholesterol in human blood. [K] / [CO3]

19. Name two laboratory diagnostic methods for oral thrush. [K] / [CO4]

20. List three common symptoms of pulmonary tuberculosis. [K] / [CO4]

21. Explain the role of 24-hour urinary free cortisol in adrenal assessment [U] / [CO2, CO3]

22. State three preventive measures of cholera. [K] / [CO4]

[3x10 = 30]

Part C

Short Essay Questions

Answer any **6** questions

Each question carries **5** marks

23. Explain complications of RSV infection and their clinical significance.	[K] / [CO5]
24. Discuss the complications and prognosis of typhoid fever	[K] / [CO4]
25. What are the normal ranges of fasting and postprandial blood sugar, and how are they altered in diabetes mellitus?	[U] / [CO3]
26. Explain the symptoms of diabetes mellitus in detail.	[K] / [CO1]
27. what are five functions of calcitonin secreted by the thyroid gland.	[K] / [CO2, CO3]
28. Explain the complications of invasive aspergillosis.	[U] / [CO5]
29. Discuss the functions of the anterior pituitary hormones with examples.	[U] / [CO2, CO3]
30. Discuss the role of stress and lifestyle in cancer progression.	[U] / [CO1]

[5x6 = 30]